

BOEKBESPREKINGS/BOOK REVIEWS

Introduction to Probability and Statistics, Part I : Probability

by Narayan C. Giri

First edition, 260 pages, Marcel Dekker, Inc., New York, 1974. \$13.75.

The title of this text and the description on the cover "...ideal textbook for undergraduate students in ... social science, economics, agriculture, ..." are a little misleading. This is a fairly rigorous text on mathematical statistics, requiring a good knowledge of calculus (for example the Borel-Cantelli lemma as proved in Chapter 4) and vector and matrix algebra. Some important results on the latter topic are neatly presented in an appendix.

The author's efforts in Chapter 1 to emphasize the role of statistics in the philosophy of scientific research are laudable. However, further examples on stochastic models, in addition to the customarily quoted coin flipping and sex-of-offspring cases, would probably be appreciated by students.

A rigorous approach (perhaps overly so) to the definition of probability occupies most of the second chapter. The short section on pairwise independence, a topic not often mentioned in texts of this nature, is commendable. The third chapter presents standard material on random variables, probability distributions, and generating functions, while the fourth chapter is concerned with a series of results on stochastic convergence and on limit theorems. The final chapter proceeds at a rapid pace through the theory of the standard univariate distributions and the multinomial and multivariate normal distributions. Some of the material on the multivariate normal distribution is a little clumsily presented. One of the most important properties of normal variates is that zero correlation is necessary and sufficient for independence. This result is not clearly presented; the necessity is discussed under the definition of the covariance matrix and the sufficiency is proved two pages later. Further, three theorems are required to prove that, if X has normal multivariate distribution, then AX , where A is a matrix of full row-rank, also has normal multivariate distribution. A short section on sampling distributions, which includes a brief mention of the Wishart distribution and a rather dated version of Cochran's Theorem, concludes the chapter.

A series of exercises is presented in each chapter and a brief set of statistical tables is published in an appendix. The book features the modern, cheaper method of photo-reproduction (of excellent quality) from a typed manuscript. The text is sprinkled with trivial typographical errors, in particular the underlining of capital letters representing vectors as opposed to scalars or matrices is often absent.

Most of the material covered would be taught in parts of first and second courses

for student majoring in statistics in South Africa. However, as a result of the cryptic style — series of definitions, theorems, exercises — and the absence of any application of probability theory to statistical inference, it would be necessary to use this book in conjunction with a second text. Perhaps this is the objective in publishing in two parts.

R.M. PRINGLE
University of Natal

Advances in Probability and Related Topics

Edited by P. Ney and S. Port

403 pages. Marcel Dekker, Inc. New York, 1974. \$25.75

This volume consists of four papers, two of which are surveys of existing literature. We proceed to discuss each in turn.

1. *Multiplicative Operator Functionals and their Asymptotic Properties* by M.A. Pinsky:

After giving a rigorous definition of the Multiplicative Operator Functional (MOF) and the associated expectation semigroup, the author illustrates these concepts in the case of finite Markov chains and the Wiener process. In particular a characterization of the MOF of a Markov chain is given as a solution of a linear integral equation.

Next, a representation of a continuous MOF of a regular step process (i.e. a jump Markov process) in terms of a certain semigroup of operators is given. This result is also generalized to the discontinuous case. The last section of the paper is concerned with the representation of MOF's of a diffusion process, although most of the proofs are given only in the case of a Wiener process.

In his introduction, the author states that the purpose of the paper is "... to give the known Mathematical theory ..." of MOF's. In our opinion he has been successful in this and his results should be accessible to anyone with a solid background e.g. at the level of Gikhman and Skorokhod (1969).

2. *Sample functions of stochastic processes with stationary, independent increments* by B. Fristedt:

The sheer length of this paper (143 pages), the wealth of results contained therein, and reasons of space, precludes a thorough discussion of it here. Suffice it to say that the author's main purpose has been to present the known results of this field in an organized manner and to draw connections wherever possible. This is a paper written for experts. Anyone who has mastered the basics of the subject (e.g. Gikhman and Skorokhod (1969), especially chapter 6) and turns to this paper for a closer look, should be prepared for hard work.

3. Aspects of Random Measures and Point Processes by P. Jagers:

This paper gives an abstract treatment of point processes, based on the idea of a random measure. Random measures on a locally compact second countable space are considered. Using a generalized version of Kolmogorov's existence theorem, the existence of a Probability measure with given "marginals" on the space of random measures, is proved. Various results pertaining to the weak convergence of sequences of random measures are also derived, including one that gives a connection with weak convergence theory in function spaces.

A point process is defined as an almost surely integer-valued random measure, and particular attention is paid to the Poisson point process. Many properties of the "classical" Poisson process are derived in this more general setting. The paper concludes with a result giving a necessary and sufficient condition for convergence of sums of independent point processes to a Poisson process.

The author has spelled out his proofs in considerable detail and this paper could certainly be used as a first orientation by someone who wishes to learn more about the abstract theory of Point processes.

4. Some new stochastic integrals and Stieltjes integrals Part II. Nigh-Martingales by L.C. Young:

In this paper, integrals with both integrand and integrator stochastic, are considered. The author is mainly concerned with weakening the classical orthogonality (martingale) assumption on the integrator. One consequence is that integrals with non-random integrators reduce to special cases of the general integral – a decided improvement on the situation as it existed, where, in the author's words: "... the word stochastic has come to mean the very antithesis of deterministic, instead of its generalization".

The paper also contains an interesting discussion on the structure of probability spaces as well as an appendix giving some connections between the methods of the paper and various unsolved problems in analysis.

REFERENCE

GIKHMAN, I.I. and A.V. SKOROKHOD (1969): Introduction to the theory of random processes. W.B. Saunders Company.

F. LOMBARD
Rand Afrikaans University
